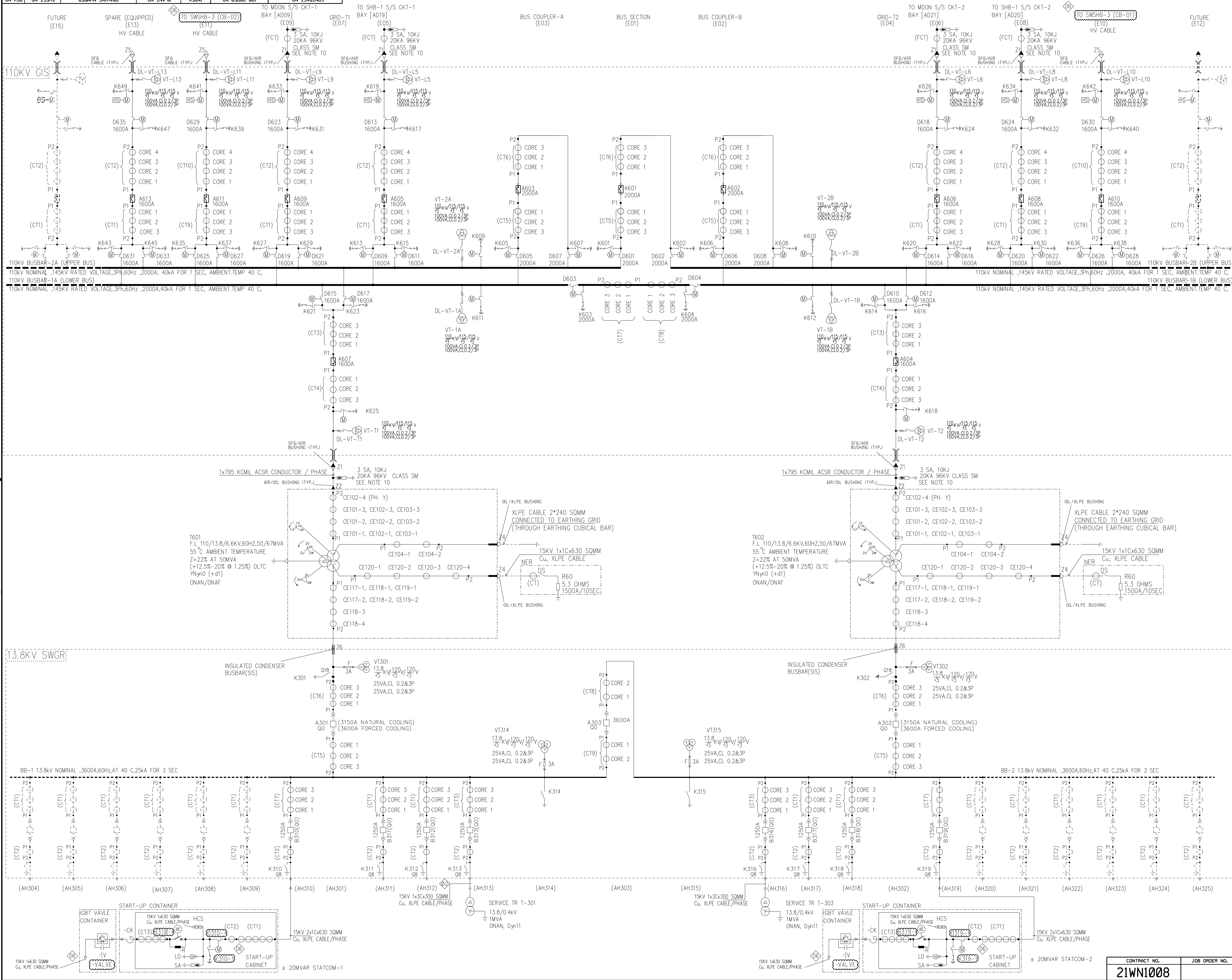




NOTES

1. ANY DEVICE, PROTECTION AS WELL AS CONTROL & STATION LEVEL DEVICES SHALL BE CONNECTED TO ETHERNET USING DUAL PORT WITH DOUBLE PARALLEL REDUNDANT COMMUNICATION SYSTEM.

2. CT/VT DATA SHALL BE PROVIDED AS PER APPROVED CALCULATIONS.

3. 13.8kV PLC APTS TO OPERATE ANY TWO TR IN PARALLEL. IN EVENT OF ANY POWER TRANSFORMER FAILURE THE REMAINING TWO POWER TRANSFORMERS SHALL OPERATE IN PARALLEL TO FEED THE ENTIRE SUBSTATION LOAD WITHOUT ANY INTERRUPTION OF SUPPLY.

4. HV & MV CABLES WILL BE SUBJECTED TO CABLE SIZING CALCULATION APPROVAL.

5. 110kV GIS CT SHALL BE BOTH SIDE AT CB AS PER PITS SLD#WT-926969

6. SPARE BAY WILL BE EQUIPPED WITH INSTALLATIONS.

7. ALL THE EQUIPMENT/RATINGS SHALL BE PROVIDED AS PER FINAL APPROVED DESIGN

8. BOT FOR WT1 (40MVA) BOT FOR WT2 ARE TENTATIVE AND TO BE CONFIRMED WITH TRANSFORMER SUPPLIER

9. THIS SLD AS PER PITS #021WN1008&PITS SLD#WT926969 AND SUPPLIER INPUT DATA

10. SA WILL BE PART OF POWER TRANS. EQUIPMENT.

11. DESIGNATION/NUMBERING OF SUBSTATION EQUIPMENT'S SHALL BE REVIEWED/FINALIZED BY OPERATION AND CONTROL (WRCC)

MAJOR EQUIPMENT SUPPLIERS

EQUIPMENT	SUPPLIER	COUNTRY OF ORIGIN
110kV GIS	HYUNDAI ELECTRIC	SOUTH-KOREA
POWER TRANSFORMER	SPIC	KSA
13.8kV SWGR	ALFAVAR	KSA
RELAY & CONTROL PANELS	HITACHI ENERGY	KSA
TELECOMMUNICATION PANELS	HITACHI ENERGY	KSA
SAS PANELS	HITACHI ENERGY	KSA
AMI	HITACHI ENERGY	KSA

LEGEND

110kV GIS TECHNICAL DATA

NOMINAL SYSTEM VOLTAGE	110 kV
RATED VOLTAGE	145 kV
RATED FREQUENCY	60Hz
RATED SHORT TIME WITHSTAND CURRENT	40 KA R.M.S. FOR 1 SEC.
IMPULSE WITHSTAND VOLTAGE - TO EARTH AND BETWEEN PHASES	650 kV
BETWEEN OPEN ISOLATOR CONTACTS	750 kV
POWER FREQUENCY WITHSTAND VOLTAGE ACROSS ISOLATING DISTANCE	315(kV)

13.8kV SWGR TECHNICAL DATA

NOMINAL VOLTAGE	13.8 kV
RATED FREQUENCY	60Hz
RATED SHORT TIME WITHSTAND CURRENT	40 KA R.M.S. FOR 3 SEC.
MAXIMUM AMBIENT TEMPERATURE	40°C

110kV GIS & 13.8kV SWGR EQUIPMENTS

SYMBOL	DESIGNATION	LOGICAL NODE	DESCRIPTION
	T601, T602	YPTR	POWER TRANSFORMER
	A6XX	XCBR	110kV CIRCUIT BREAKER
			MOTOR SPRING CHARGED OPERATION
	K6XX	XSWI	110kV HIGH SPEED EARTH SWITCH (MOTOR DRIVEN SPRING CHARGED OPERATION)
	D6XX/A6XX	XSWI	110kV COMBINED WITH DS & ES (MOTOR DRIVEN OPERATION)
	K6XX	----	110kV MAINTENANCE EARTH SWITCH (MOTOR DRIVE OPERATION)
	K3XX	XSWI	110kV MAINTENANCE EARTH SWITCH WITH MANUAL OPERATION
	R60	----	NEUTRAL EARTHING RESISTOR (NERI)
	VT-LX/VT-TX	XSWI	110kV GIS VOLTAGE TRANSFORMER (MANUAL)
	A3XX	XCBR	13.8kV SWGR CIRCUIT BREAKER
	----	ICTR	CURRENT TRANSFORMER
	VT(3XX)	TVTR	13.8kV SWGR VOLTAGE TRANSFORMER WITH DRAWABLE FUSE(HAND)
	T7	TVTR	INDUCTIVE TYPE VOLTAGE TRANSFORMER
	AUX-1, AUX-2	----	AUXILIARY TRANSFORMER
	GIB	----	SURGE ARRESTER
	Z1	----	110kV GIS GIB WITH SF6/AIR BUSHING
	-A51	----	CAPACITIVE VOLTAGE INDICATOR
	22	----	POWER TRANSFORMER'S HV AIR/OIL BUSHING
	24	----	POWER TRANSFORMER'S HV OIL/XLPE BUSHING
	25	----	110kV GIS SF6/CABLE
	26	----	INSULATED CONDENSER BUSBAR(SIS)

REFERENCE DRAWINGS

DRAWING TITLE	DRAWING NO.
DRAWING CONTROL SHEET	WE-0668661
PITS FOR CONSTRUCTION OF SHOABA-3 110/13.8kV SUBSTATION	PITS-21WN1008

AS-BUILT

BY

DATE

06	14.11.2024	ABA	REVISED AS PER COMMENTS	F.M	T.TH	T.S
05	28.08.2023	ABA	REVISED AS PER COMMENTS	F.M	T.TH	T.S
04	28.08.2023	ABA	REVISED AS PER COMMENTS	F.M	T.TH	T.S
03	19.03.2023	ABA	REVISED AS PER COMMENTS	F.M	T.TH	T.S
02	01.01.2023	ABA	REVISED AS PER COMMENTS	F.M	T.TH	T.S

REV.

DATE

BY

DESCRIPTION

CHKD.

CERT.

APPD.

DRAWING COMPLETION CERTIFICATE (DCC)

الشركة الوطنية لنقل الكهرباء

نقل الكهرباء

National Grid SA

DRAWN BY

ABASHRAF

CHECKED BY

F.MOHAMED

SCALE

DATE STARTED

15.08.2022

DATE COMPLETED

OPRG. DEPT.

ENG. DEPT.

THIS DRAWING IS NOT TO BE USED FOR CONSTRUCTION OR FOR ORDERING MATERIAL UNTIL CERTIFIED AND DATED

APPROVAL / CERTIFICATION INFORMATION

DOC. NO.

REV. NO.

MAIN ONE LINE DIAGRAM

POWER SUPPLY FOR DESALINATION PROJECT

MK-SWPC

SHOABA-MAKKAH3

SAUDI ARABIA

CONTRACT NO.

21WN1008

JOB ORDER NO.

INDEX

P

PLANT NO.

J754

DRAWING NUMBER

WA-0668663

SHEET NO.

01 of 03

REV. NO.

05

NG-A (2012/01)

DMS

VECTOR

NGA08

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